# LAUGHING LUMI

## Evidence-Led Child–AI Interaction and Product Validation

An Investor Research Report

October 2026


## ABSTRACT

Early-childhood learning applications face a distinctive product challenge: a system must not merely recognize a child's speech but deliver an instruction the child can understand, act upon and recover from when misunderstandings occur. This research report synthesizes peer-reviewed findings on children's voice interaction, developmentally appropriate digital media, multimedia learning and parents' evaluation of educational applications. It proposes an evidence-led product strategy for Laughing Lumi, beginning with voice conversation and literacy-accessible screen design, followed by computer vision and broader play-and-learning activities. It also specifies a measurable approach to usability testing, a brief caregiver questionnaire and subsequent product optimization. Published findings are distinguished from proposed Lumi experiments throughout; no claim of verified Lumi learning outcomes, user retention or commercial conversion is made.



## KEYWORDS

Child–AI interaction; conversational interfaces; early literacy; educational technology; speech recognition; product validation; caregiver experience; evidence-based design.


## RESEARCH HIGHLIGHTS

- Published child–voice interaction research identifies a gap between recognized speech and appropriate conversational response.
- Literature on digital multimedia indicates design-dependent educational effects: useful visual/auditory scaffolds can help, while unnecessary interactions may distract.
- Parents' stated educational app choices respond to both evidence-oriented product cues and social proof; these are not substitutes for observed retention or purchases.
- The proposed product sequence reduces technical and research uncertainty by validating voice interaction and accessible UI before camera-based perception.
- A repeatable evaluation design can connect interaction quality, caregiver appraisal, meaningful repeated use and—later—commercial evidence.

# 1 | INTRODUCTION: THE PRODUCT RESEARCH QUESTION

Young children may engage with spoken instructions before they can independently read text or formulate adult-like questions. This creates an opportunity for voice-first play interfaces, but it also creates a reliability problem: the system must correctly interpret variable child speech, maintain a coherent interaction and present language that is actionable rather than merely engaging.

The motivating concern—that Lumi's dialogue may be too linguistically complex for younger users—was identified by the co-founder as a design hypothesis. The corresponding research question is not “How can we make AI sound simpler?” but “Which combinations of spoken instructions, contextual visual cues and repair strategies enable children to complete intended activities reliably, with an age- and language-appropriate experience?”

Evidence reviewed here is relevant to this hypothesis; it does not establish that Laughing Lumi currently suffers from any measured level of misunderstanding. Likewise, the review does not assert a universal ideal word count or a deterministic relationship between age and sentence length.

# 2 | METHODS AND EVIDENCE BASE

This document is a structured, purposive synthesis of peer-reviewed empirical research, a child-rights policy framework and applied measurement literature. Sources were selected for direct relevance to four investment-relevant questions: (i) whether child-directed voice interaction has unresolved usability challenges; (ii) whether multimodal scaffolding and early-literacy adaptation are plausible design responses; (iii) which experience factors parents report as valuable; and (iv) how product experiments could produce comparable evidence over time. Publication records, original abstracts and—when accessible—full text were checked for study context, sample and key result. This is not a registered systematic review, and its selected studies do not estimate the effect of Laughing Lumi.

Evidence categories used in this report:

PUBLISHED EMPIRICAL FINDING — source results observed in another study;

RESEARCH-INFORMED DESIGN HYPOTHESIS — a plausible Lumi design option requiring testing;

PROPOSED MEASUREMENT — defined data instrument not yet implemented or observed in Lumi.

## 2.1 | CHILDREN'S VOICE ASSISTANTS: RECOGNITION IS NOT ENOUGH

Kim et al. (2022) observed 28 children aged 5–10 interacting with a commercial voice assistant. The authors reported approximately 84% accuracy in transcribing unscripted children's speech while appropriate responses to the content occurred about half the time. The two numbers measure distinct outcomes and must not be treated as a direct head-to-head test of competing

models. Nevertheless, they illustrate why speech transcription and conversational appropriateness require independent quality criteria.

Yuan et al. (2019), studying 87 children aged 5–12 and 27 adults across three Wizard-of-Oz interfaces, observed that children often needed help to reformulate questions effectively. This highlights a design opportunity for supportive clarification rather than repeatedly expecting the child to rephrase in a manner convenient to the model.

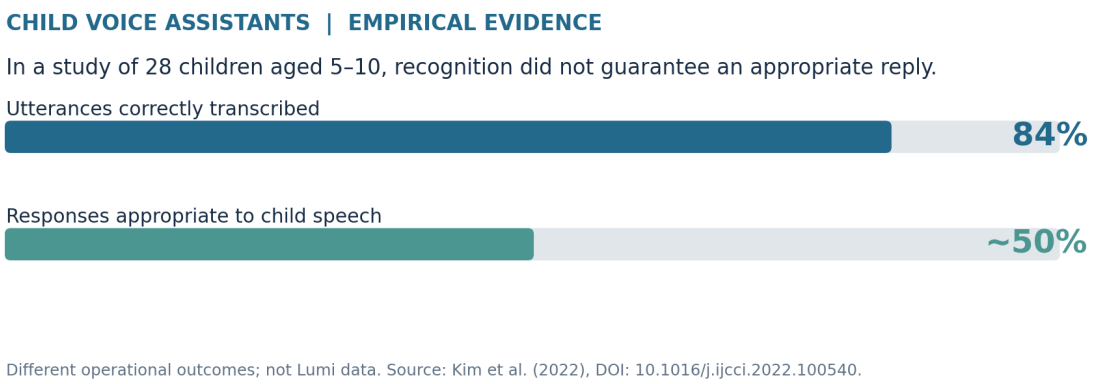



*Figure 1. Results reported for a commercial voice assistant used by children; source: Kim et al. (2022). The two observed metrics have different definitions and are not Lumi performance measurements.*

## 2.2 | INSTRUCTION COMPLEXITY AND EMERGENT LITERACY

Jaroslawska et al. (2016) experimentally investigated instruction-following and action enactment among children aged 7–9. Their evidence supports testing the roles of action, memory demands and cueing; it does not justify fixed sentence-length thresholds for preschool users. Whitehurst and Lonigan (1998) describe emergent literacy as multidimensional precursors to later reading, supporting a distinction among chronological age, reading readiness and spoken-language proficiency.

For Lumi, this motivates a spoken-instruction design in which critical referents are concrete and available in the child's environment, the action is identifiable, and multiple independent tasks are not casually combined in a long turn. On-screen text should be complementary rather than the sole channel for a child who is not yet reading. A somewhat longer instruction accompanied by a useful visual cue may be easier to follow than a short but referentially ambiguous one; this proposition must be measured rather than assumed.

## 2.3 | MULTIMEDIA LEARNING: POSITIVE EFFECTS DEPEND ON DESIGN

Takacs et al. (2015) conducted a meta-analysis of technology-enhanced storybooks involving 43 studies and 2,147 children. Compared with more traditional storytelling, the research estimated small additional benefits for story comprehension (Hedges' g = 0.17) and expressive vocabulary (g = 0.20). The same synthesis cautioned that interactive hotspots and games could distract, whereas congruent multimedia elements could be beneficial.

These results support a disciplined design thesis: visual support should advance the intended task, not merely make a screen more animated. They do not show that conversational AI by itself produces vocabulary gains or that Lumi has demonstrated learning efficacy.

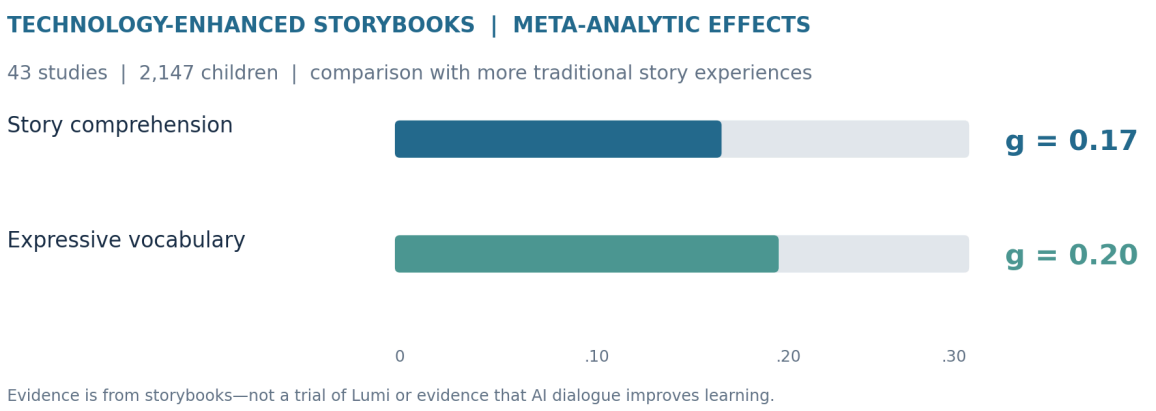



*Figure 2. Published standardized effects for technology-enhanced storybooks, not results for Laughing Lumi. Source: Takacs et al. (2015).*

## 2.4 | WHAT MAKES AN EDUCATIONAL APPLICATION VALUABLE TO PARENTS?

Parent choice is a distinct commercial research question, separate from children's comprehension. Montazami et al. (2022) observed parent responses to ten mock educational app pages and found evidence-oriented design cues valued more than generic promotional language within the study. In a later study, Liptrot et al. (2025) reported that 149 parents evaluating 18 researcher-created math app pages were strongly influenced by positive ratings and reviews; the additional role of explicit educational-quality benchmarks was less clear.

These findings are complementary rather than contradictory: educational substance may be needed to deliver long-term value, while external trust cues can influence initial selection. They are observations of stated choices in simulated storefronts, not actual Lumi downloads or purchases.

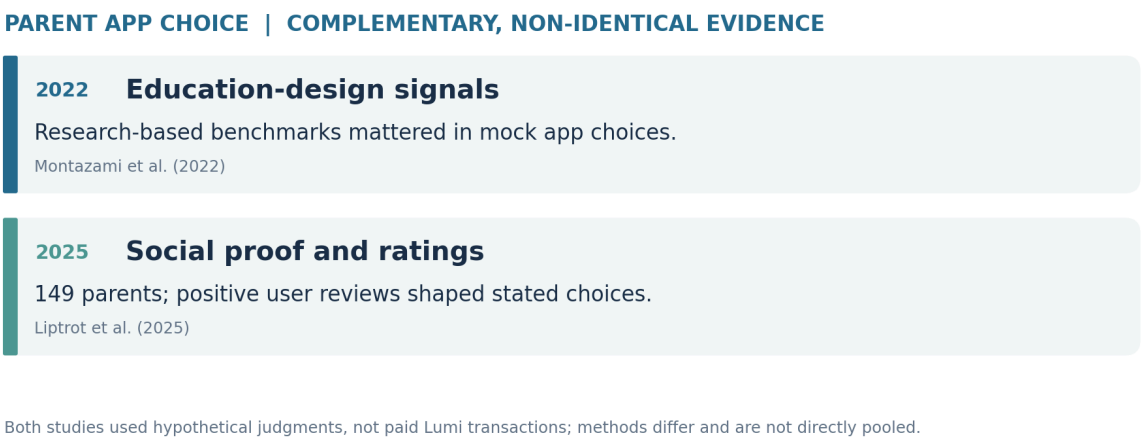



*Figure 3. Two published studies address distinct parts of parental app evaluation. The studies differ in manipulation and sample and are not statistically pooled.*

## 3 | PRODUCT STRATEGY: TECHNICAL SEQUENCING THAT CAN BE VALIDATED

The proposed research-led development order is deliberate. Phase 1 prioritizes voice conversation as the core functional experience, together with developmentally appropriate visual UI and accessible alternatives to text input. Phase 2 examines computer vision only where it adds observable value to a child's real-world activity. Phase 3 broadens the portfolio of bounded play-and-learning experiences and tests language or learning adaptation when supporting evidence exists.

This order is a product strategy, not a claim that each stage is already implemented. It makes it possible to study one consequential design question at a time, to observe relevant operating costs and to avoid interpreting problems from a future perception system as problems in an unstable underlying voice interface.




*Figure 4. Proposed stage sequence. Reading-accessible visual design is included in the voice phase; camera-based computer vision is a separate later capability.*

### 3.1 | PHASE 1: SPOKEN DIALOGUE AND SCREEN-READING ACCESSIBILITY

The minimum viable interaction is defined by five testable requirements: the system delivers audible instructions reliably; the intended action and target can be identified; uncertain child speech results in a neutral clarification or accessible fallback; children can stop or ask an adult for help; and the interface does not require independent reading or typing to complete tasks intended for pre-readers. Unlike a universal word-count rule, these outcomes can be operationalized and directly assessed.

Screen text can be configured as illustrative development modes: icon-and-audio-led experiences for nonreaders; short, synchronized caption-and-audio cues for emerging readers; and optional expanded text for children with more independent reading ability. These are research candidates, not validated age brackets. Caregiver-reported reading readiness and actual task context should supplement chronological age.

### 3.2 | PHASE 2: CAMERA-BASED UNDERSTANDING OF REAL-WORLD ACTION

Once voice interaction is reliable enough to support controlled experiences, computer vision can be tested in bounded, meaningful tasks: observing whether a requested object or simple physical action is present, or providing multimodal feedback about real-world play. This research phase requires independently annotated reference data, false-positive and false-negative analysis, latency measurement, child-data minimization and a noncamera interaction route. There are no verified Lumi computer-vision accuracy values in this report.

### 3.3 | PHASE 3: EXTENDED ACTIVITIES AND INTERNATIONAL ADAPTATION

MOVE, LISTEN, CREATE and DO represent useful conceptual categories for subsequent feature validation. The commercial value of each should be assessed using a defined child activity, the caregiver's job-to-be-done, repeat-use motivation, safe execution, technical cost and testable educational objective. A China-facing English-through-play variant could use listening, action and visual scaffolding, but it is a distinct product and market hypothesis. Neither English-learning effectiveness nor China-market demand follows automatically from the general literature.

## 4 | VALIDATION METHOD: TURNING PRODUCT DECISIONS INTO EMPIRICAL QUESTIONS

The first controlled experiment concerns child-directed speech rather than overall “AI intelligence.” Within a clearly specified game task, three matched versions can be compared:

CONDITION A — actual, version-identified current dialogue;

CONDITION B — structured spoken instruction with an explicit action and referent;

CONDITION C — the same spoken instruction as B combined with a congruent visual cue.

If a current app build is unavailable, a reconstructed A condition must be identified as a mock comparison rather than a measured production baseline. Use identical targets, content objectives and task contexts, with counterbalancing of matched items. Only after relevant technical, privacy, ethics and guardian approval conditions are met should a supervised study include children.




*Figure 5. Proposed comparison design. Conditions are not reported as completed experimental arms.*

The primary outcome is independently observed correct task action without adult assistance, among trials in which the task was properly delivered. Technical invalidity, child opt-out, no response, caregiver help and safety interruptions should be coded separately. Secondary outcomes include response time, clarification rate, repair success, speech recognition and playback failure, repeat requests and difficulty with screen-based input. Repeated trials from the same child are not independent observations; analysis should account for participant and task-item effects when the design and sample permit.

A second experiment, following adequate voice stability, varies text density, pictorial cues and narration while holding task objectives constant. Its purpose is to measure accessibility and usability, not to infer literacy development from a single session. Actual educational learning must be evaluated separately using baseline competence, matched comparisons and appropriate follow-up tasks.

## 5 | A SHORT CAREGIVER MEASUREMENT INSTRUMENT

Caregivers are the appropriate initial respondents for a brief, optional post-session experience survey; young children should not be forced to read a questionnaire in order to continue play. The proposed six-item scale draws its *construct selection* from literature on perceived usability, engagement, learning, concerns and value, including the PEAU-p study of 435 parent respondents (Vaiopoulou et al., 2021). PEAU-p was a validated 27-item, six-factor instrument in its original setting. The six proposed Lumi items are newly drafted indicators, not a validated abbreviated version of PEAU-p.

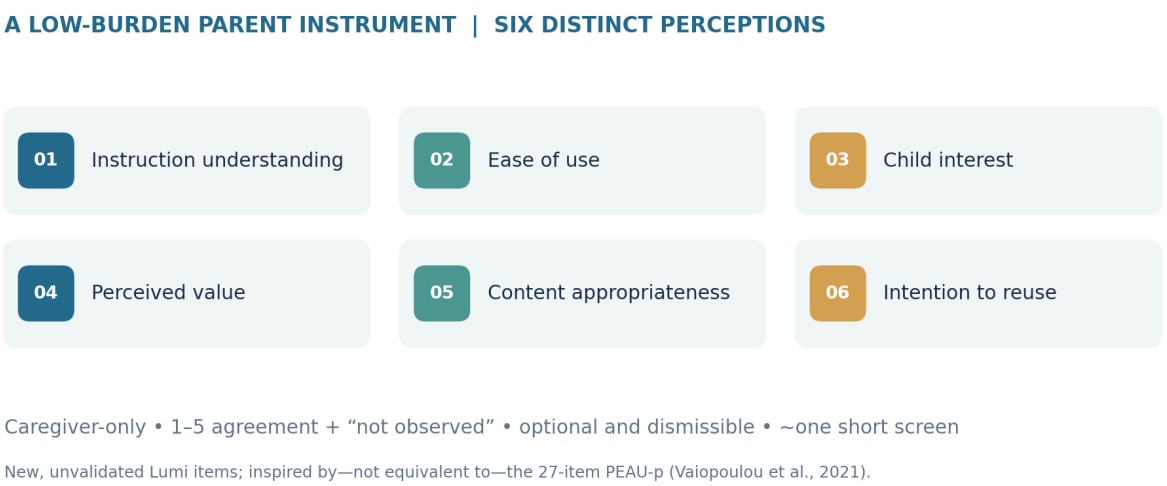



*Figure 6. Six proposed caregiver-report dimensions. These describe impressions of a particular session, not observed educational efficacy or demonstrated payment behavior.*

## 5.1 | EXACT MICRO-SURVEY WORDING

Respondent instructions: “Thinking about the activity you just observed, how much do you agree with each statement?”

Q1 — “My child seemed to understand Lumi's spoken instructions.”

Q2 — “My child could use the activity with an appropriate amount of help.”

Q3 — “My child seemed interested in the activity.”

Q4 — “I felt this activity was worthwhile for my child.”

Q5 — “The content of this activity seemed appropriate for my child.”

Q6 — “I would choose a similar Lumi activity for my child again.”

Response format: 1 strongly disagree; 2 disagree; 3 neither agree nor disagree; 4 agree; 5 strongly agree. “Did not observe / not applicable” and a skip option are independent missing-response codes. A small optional difficulty category question can identify a concrete usability problem. The form should be shown only to a caregiver after a meaningful interaction and should remain dismissible.

An initial cognitive-interview round should verify item interpretation, burden and cross-language equivalence. Because the items measure different constructs, they should not automatically be summed into an overall psychometric score or assigned the published PEAU-p reliability coefficients.

# 6 | FROM MEASUREMENT TO INVESTMENT-RELEVANT PRODUCT EVIDENCE

Product validation becomes commercially useful when it separates five increasingly demanding observations: (i) a reliable interaction was delivered, (ii) the child understood the task, (iii) the caregiver perceived value, (iv) the family voluntarily returned for meaningful use, and (v) there was separately observed willingness to purchase or actual paid conversion. Each transition requires its own evidence. Positive survey ratings do not independently establish either learning or recurring revenue.

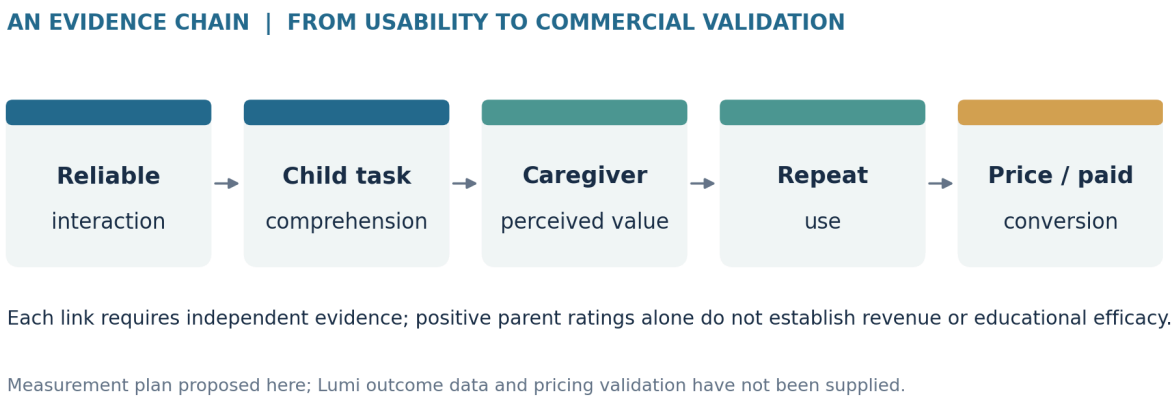



*Figure 7. Proposed evidence-building sequence. No Lumi outcomes are reported here.*

## 6.1 | PRODUCT METRICS AND INTERPRETATION

Technical quality: percentage of eligible prompts with audible playback; median and upper-tail time to first speech; separately verified ASR transcription and intent correctness; repair outcomes; severity-classified child-safety events.

Child comprehension: correctly performed independent actions / technically valid task trials, reported alongside number of unique participating children and invalid trials.

Caregiver appraisal: per-item ordinal response distribution, median, proportion rating 4–5, number shown the form, completed responses and nonresponse; six items remain separate.

Meaningful return: unique eligible activated households with a further qualifying activity in a prespecified D7 or D30 window / eligible, fully observable activation cohort; distinguish mere app launches from completed activity.

Later commercial validation: properly sourced actual trial offers, willingness-to-pay research and paid purchases; maintain independent denominators and country/billing definitions.

No real Lumi values for these indicators have been supplied for this report. A professional external dashboard should therefore display “measurement pending” rather than illustrative percentages that resemble actual performance.

### 6.2 | INSTRUMENTATION AND REPRODUCIBILITY

A lean event model records the versioned voice/UI condition, session and task identifiers, task delivered, speech playback, attempted action, independent outcome, technical failure class, caregiver help and questionnaire invitation/completion. Minimize personal information; children's audio and video must not be collected by default. All applicable child-data, caregiver authorization, safety and research-ethics requirements must be resolved for the specific study before sensitive data are collected.

For survey item j, a positive-response percentage is computed as valid ratings of 4 or 5 divided by valid responses of 1–5 to that item. For an A/B task comparison, provide child count and task count, observed arm-level proportions and an estimate of the between-condition difference with uncertainty. Statistical analysis should account for repeated child/item measurements and missingness; a small exploratory study primarily evaluates feasibility rather than proving a product-level causal effect.

## 7 | AN ITERATIVE PRODUCT RESEARCH ENGINE

Rather than a one-off demonstration, the recommended operating model is a reproducible cycle: identify a failure or opportunity; formalize a testable hypothesis; review relevant research; create and version one product variant; conduct approved technical and user testing; combine task logs with caregiver reports; quantify changes and uncertainty; decide whether to retain, revise, retest or revert; and monitor the next version.

The practical commercial benefit is cumulative learning. A research program that continuously identifies why children misunderstand a task, how caregivers experience it and whether families return can improve product decisions while reducing the temptation to expand functionality before core interaction quality is understood. The aim is not to manufacture favorable scores; it is to make investment, engineering and design decisions based on auditable evidence.

## 8 | DISCUSSION: AN INVESTABLE HYPOTHESIS, NOT A CLAIM OF PROVEN OUTCOMES

The published evidence provides a coherent rationale for a voice-first, literacy-aware and measurement-led approach to children's AI play. First, recognizing speech and producing an appropriate conversational response are meaningfully different engineering problems. Second, instructional memory, emergent literacy and congruent multimedia support explain why young children's interfaces require specific design rather than an adult chat experience with a playful voice. Third, parents' app selection is influenced by both perceived educational substance and market trust signals. These observations support a plausible product-development thesis.

They do not establish that Lumi has already solved these problems, that users will subscribe at any particular price, that visual AI will improve learning, or that early usability improvements will

translate into durable retention. These outcomes belong to the next empirical studies and market-validation work.

## 9 | CONCLUSION

Laughing Lumi's proposed differentiation is not unrestricted child conversation alone. It is the combination of bounded child-appropriate voice interaction, an interface usable before independent reading, progressively added real-world activity understanding and a repeatable evaluation methodology. The first measurable milestone is a reliable voice-and-interface experience in which children can understand and complete tasks safely; later milestones require appropriately designed education and market studies. This staged approach connects a defensible research hypothesis to a practical technical roadmap without overstating demonstrated performance.

## REFERENCES | APA 7


American Educational Research Association, American Psychological Association, & National Council on Measurement in Education. (2014). Standards for educational and psychological testing. AERA. https://ncme.org/resources/books/testing-standards/

Jaroslawska, A. J., Gathercole, S. E., Allen, R. J., & Holmes, J. (2016). Following instructions from working memory: Why does action at encoding and recall help? Memory & Cognition, 44, 1183–1191. https://doi.org/10.3758/s13421-016-0636-5

Kim, M. K., Druga, S., Esmaeili, S., Woodward, J., Shaw, A., Jain, A., Langham, J., Hollingshead, K., Lovato, S. B., Beneteau, E., Ruiz, J., Anthony, L., & Hiniker, A. (2022). Examining voice assistants in the context of children's speech. International Journal of Child-Computer Interaction, 34, Article 100540. https://doi.org/10.1016/j.ijcci.2022.100540

Liptrot, E., Montazami, A., Pearson, H. A., & Dubé, A. K. (2025). Why this app: How user ratings and app store rankings impact parents' choice of educational apps. Computers & Education, 238, Article 105410. https://doi.org/10.1016/j.compedu.2025.105410

Montazami, A., Pearson, H. A., Dubé, A. K., Kacmaz, G., Wen, R., & Alam, S. S. (2022). Why this app? How parents choose good educational apps from app stores. British Journal of Educational Technology, 53(6), 1766–1792. https://doi.org/10.1111/bjet.13213

National Association for the Education of Young Children, & Fred Rogers Center. (2012). Technology and interactive media as tools in early childhood programs serving children from birth through age 8 [Position statement]. https://www.naeyc.org/sites/default/files/globally-shared/downloads/PDFs/resources/topics/ps_technology.pdf

Rodden, K., Hutchinson, H., & Fu, X. (2010). Measuring the user experience on a large scale: User-centered metrics for web applications. Proceedings of the SIGCHI Conference on Human Factors in Computing Systems. https://doi.org/10.1145/1753326.1753687

Takacs, Z. K., Swart, E. K., & Bus, A. G. (2015). Benefits and pitfalls of multimedia and interactive features in technology-enhanced storybooks: A meta-analysis. Review of Educational Research, 85(4), 698–739. https://doi.org/10.3102/0034654314566989

UNICEF Innocenti. (2025). Guidance on AI and children (Version 3.0). UNICEF. https://www.unicef.org/innocenti/reports/policy-guidance-ai-children

Vaiopoulou, J., Papadakis, S., Sifaki, E., Stamovlasis, D., & Kalogiannakis, M. (2021). Parents' perceptions of educational apps use for kindergarten children: Development and validation of a new instrument (PEAU-p) and exploration of parents' profiles. Behavioral Sciences, 11(6), Article 82. https://doi.org/10.3390/bs11060082

Whitehurst, G. J., & Lonigan, C. J. (1998). Child development and emergent literacy. Child Development, 69(3), 848–872. https://doi.org/10.1111/j.1467-8624.1998.tb06247.x

Yuan, Y., Thompson, S., Watson, K., Chase, A., Senthilkumar, A., Bernheim Brush, A. J., & Yarosh, S. (2019). Speech interface reformulations and voice assistant personification preferences of children and parents. International Journal of Child-Computer Interaction, 21, 77–88. https://doi.org/10.1016/j.ijcci.2019.04.005

## RESEARCH SCOPE NOTE

This report is an evidence-based product research prospectus. References describe published studies, while all planned Lumi experiments and metrics remain unobserved until performed. It contains no verified company revenue, user base, child outcome, model accuracy or subscription forecasts.